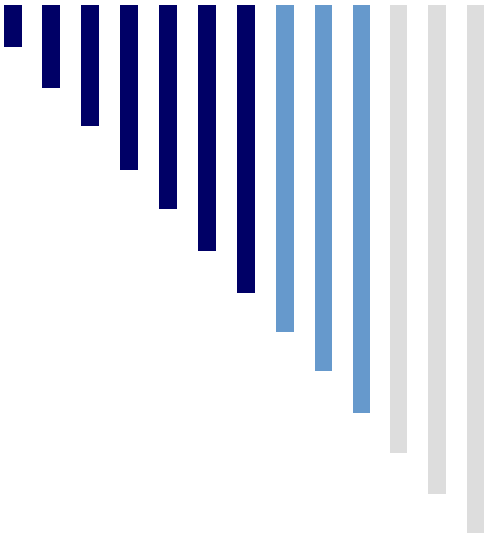


Confidence Interval Estimation
&
Hypothesis Testing

Problems

8.9 A bottled water distributor wants to estimate the amount of water contained in 1-gallon bottles purchased from a nationally known water bottling company. The water bottling company's specifications state that the standard deviation of the amount of water is equal to 0.02 gallon. A random sample of 50 bottles is selected, and the sample mean amount of water per 1-gallon bottle is 0.995 gallon.

a. Construct a 99% confidence interval estimate for the population mean amount of water included in a 1-gallon bottle.

b. On the basis of these results, do you think that the distributor has a right to complain to the water bottling company? Why?

c. Must you assume that the population amount of water per bottle is normally distributed here? Explain.

d. Construct a 95% confidence interval estimate. How does this change your answer to (b)?

8.10 The operations manager at a compact fluorescent light bulb (CFL) factory needs to estimate the mean life of a large shipment of CFLs. The manufacturer's specifications are that the standard deviation is 1,000 hours. A random sample of 64 CFLs indicated a sample mean life of 7,500 hours.

a. Construct a 95% confidence interval estimate for the population mean life of compact fluorescent light bulbs in this shipment.

b. Do you think that the manufacturer has the right to state that the compact fluorescent light bulbs have a mean life of 8,000 hours? Explain.

c. Must you assume that the population compact fluorescent light bulb life is normally distributed? Explain.

d. Suppose that the standard deviation changes to 800 hours. What are your answers in (a) and (b)?

Problems

8.15 A marketing researcher wants to estimate the mean amount spent ($) on Amazon.com by Amazon Prime member shoppers. Suppose a random sample of 100 Amazon Prime member shoppers who recently made a purchase on Amazon.com yielded a mean of $1,500 and a standard deviation of $200.

a. Construct a 95% confidence interval estimate for the mean spending for all Amazon Prime member shoppers.

b. Interpret the interval constructed in (a).

8.16 A survey of nonprofit organizations showed that online fundraising has increased in the past year. Based on a random sample of 84 nonprofits, the mean one-time gift donation in the past year was $82, with a standard deviation of $9.

a. Construct a 95% confidence interval estimate for the population mean one-time gift donation.

b. Interpret the interval constructed in (a).

8.24 The file MobileCommerce contains mobile commerce penetration values, the percentage of the country population that bought something online via a mobile phone in the past month, for twenty-eight of the world's economies:

17 15 15 13 27 12 20 23 9 9

16 6 19 14 11 14 8 15 23 11

37 17 11 19 27 18 18 15

Source: Data extracted from "Digital, Social, and Mobile in 2015,"

bit.ly/1yHE8vK.

a. Construct a 95% confidence interval estimate for the population mean mobile commerce penetration.

b. What assumption do you need to make about the population to construct the interval in (a)?

c. Given the data presented, do you think the assumption needed in (a) is valid? Explain.

PROBLEMS

1. The manager of the Danvers-Hilton Resort Hotel stated that the mean guest bill for a weekend is \$600 or less. A member of the hotel's accounting staff noticed that the total charges for guest bills have been increasing in recent months. The accountant will use a sample of future weekend guest bills to test the manager's claim.
   a. Which form of the hypotheses should be used to test the manager's claim? Explain.

$$H_0: \mu \geq 600 \qquad H_0: \mu \leq 600 \qquad H_0: \mu = 600$$
$$H_a: \mu < 600 \qquad H_a: \mu > 600 \qquad H_a: \mu \neq 600$$

   b. What conclusion is appropriate when H_0 cannot be rejected?
   c. What conclusion is appropriate when H_0 can be rejected?

2. The manager of an automobile dealership is considering a new bonus plan designed to increase sales volume. Currently, the mean sales volume is 14 automobiles per month. The manager wants to conduct a research study to see whether the new bonus plan increases sales volume. To collect data on the plan, a sample of sales personnel will be allowed to sell under the new bonus plan for a one-month period.
   a. Develop the null and alternative hypotheses most appropriate for this situation.
   b. Comment on the conclusion when H_0 cannot be rejected.
   c. Comment on the conclusion when H_0 can be rejected.

3. A production line operation is designed to fill cartons with laundry detergent to a mean weight of 32 ounces. A sample of cartons is periodically selected and weighed to determine whether underfilling or overfilling is occurring. If the sample data lead to a conclusion of underfilling or overfilling, the production line will be shut down and adjusted to obtain proper filling.
   a. Formulate the null and alternative hypotheses that will help in deciding whether to shut down and adjust the production line.
   b. Comment on the conclusion and the decision when H_0 cannot be rejected.
   c. Comment on the conclusion and the decision when H_0 can be rejected.

4. Because of high production-changeover time and costs, a director of manufacturing must convince management that a proposed manufacturing method reduces costs before the new method can be implemented. The current production method operates with a mean cost of \$220 per hour. A research study will measure the cost of the new method over a sample production period.
   a. Develop the null and alternative hypotheses most appropriate for this study.
   b. Comment on the conclusion when H_0 cannot be rejected.
   c. Comment on the conclusion when H_0 can be rejected.

5. Duke Energy reported that the cost of electricity for an efficient home in a particular neighborhood of Cincinnati, Ohio, was $104 per month (*Home Energy Report,* Duke Energy, March 2012). A researcher believes that the cost of electricity for a comparable neighborhood in Chicago, Illinois, is higher. A sample of homes in this Chicago neighborhood will be taken and the sample mean monthly cost of electricity will be used to test the following null and alternative hypotheses.

$$H_0: \mu \leq 104$$
$$H_a: \mu > 104$$

a. Assume the sample data led to rejection of the null hypothesis. What would be your conclusion about the cost of electricity in the Chicago neighborhood?
b. What is the Type I error in this situation? What are the consequences of making this error?
c. What is the Type II error in this situation? What are the consequences of making this error?

6. The label on a 3-quart container of orange juice states that the orange juice contains an average of 1 gram of fat or less. Answer the following questions for a hypothesis test that could be used to test the claim on the label.
   a. Develop the appropriate null and alternative hypotheses.
   b. What is the Type I error in this situation? What are the consequences of making this error?
   c. What is the Type II error in this situation? What are the consequences of making this error?

7. Carpetland salespersons average $8000 per week in sales. Steve Contois, the firm's vice president, proposes a compensation plan with new selling incentives. Steve hopes that the results of a trial selling period will enable him to conclude that the compensation plan increases the average sales per salesperson.
   a. Develop the appropriate null and alternative hypotheses.
   b. What is the Type I error in this situation? What are the consequences of making this error?
   c. What is the Type II error in this situation? What are the consequences of making this error?

8. Suppose a new production method will be implemented if a hypothesis test supports the conclusion that the new method reduces the mean operating cost per hour.
   a. State the appropriate null and alternative hypotheses if the mean cost for the current production method is $220 per hour.
   b. What is the Type I error in this situation? What are the consequences of making this error?
   c. What is the Type II error in this situation? What are the consequences of making this error?

SELF TEST

9.14 The quality-control manager at a compact fluorescent light bulb (CFL) factory needs to determine whether the mean life of a large shipment of CFLs is equal to 7,500 hours. The population standard deviation is 1,000 hours. A random sample of 64 CFLs indicates a sample mean life of 7,250 hours.

a. At the 0.05 level of significance, is there evidence that the mean life is different from 7,500 hours?

b. Compute the *p*-value and interpret its meaning.

c. Construct a 95% confidence interval estimate of the population mean life of the CFLs.

d. Compare the results of (a) and (c). What conclusions do you reach?

9.16 A bottled water distributor wants to determine whether the mean amount of water contained in 1-gallon bottles purchased from a nationally known water bottling company is actually 1 gallon. You know from the water bottling company specifications that the standard deviation of the amount of water per bottle is 0.02 gallon. You select a random sample of 50 bottles, and the mean amount of water per 1-gallon bottle is 0.995 gallon.

a. Is there evidence that the mean amount is different from 1.0 gallon? (Use $\alpha = 0.01$.)

b. Compute the *p*-value and interpret its meaning.

c. Construct a 99% confidence interval estimate of the population mean amount of water per bottle.

d. Compare the results of (a) and (c). What conclusions do you reach?

22. CCN and ActMedia provided a television channel targeted to individuals waiting in supermarket checkout lines. The channel showed news, short features, and advertisements. The length of the program was based on the assumption that the population mean time a shopper stands in a supermarket checkout line is 8 minutes. A sample of actual waiting times will be used to test this assumption and determine whether actual mean waiting time differs from this standard.

a. Formulate the hypotheses for this application.

b. A sample of 120 shoppers showed a sample mean waiting time of 8.4 minutes. Assume a population standard deviation of $\sigma = 3.2$ minutes. What is the p-value?

c. At $\alpha = .05$, what is your conclusion?

d. Compute a 95% confidence interval for the population mean. Does it support your conclusion?

SELF TEST

9.24 You are the manager of a restaurant for a fast-food franchise. Last month, the mean waiting time at the drive-through window for branches in your geographic region, as measured from the time a customer places an order until the time the customer receives the order, was 3.7 minutes. You select a random sample of 64 orders. The sample mean waiting time is 3.57 minutes, with a sample standard deviation of 0.8 minute.

a. At the 0.05 level of significance, is there evidence that the population mean waiting time is different from 3.7 minutes?

b. Because the sample size is 64, do you need to be concerned about the shape of the population distribution when conducting the *t* test in (a)? Explain.

9.26 A marketing researcher wants to estimate the mean amount spent ($) on Amazon.com by Amazon Prime member shoppers. Suppose a random sample of 100 Amazon Prime member shoppers who recently made a purchase on Amazon.com yielded a mean amount spent of $1,500 and a standard deviation of $200.

a. Is there evidence that the population mean amount spent on Amazon.com by Amazon Prime member shoppers is different from $1,475? (Use a 0.05 level of significance.)

b. Determine the *p*-value and interpret its meaning.

9.28 The file **FastFood** contains the amount that a sample of fifteen customers spent for lunch ($) at a fast-food restaurant:

7.42	6.29	5.83	6.50	8.34	9.51	7.10	6.80	5.90
4.89	6.50	5.52	7.90	8.30	9.60			

a. At the 0.05 level of significance, is there evidence that the mean amount spent for lunch is different from $6.50?

b. Determine the *p*-value in (a) and interpret its meaning.

c. What assumption must you make about the population distribution in order to conduct the *t* test in (a) and (b)?

d. Because the sample size is 15, do you need to be concerned about the shape of the population distribution when conducting the *t* test in (a)? Explain.

11. A *Bloomberg Businessweek* subscriber study asked, "In the past 12 months, when traveling for business, what type of airline ticket did you purchase most often?" A second question asked if the type of airline ticket purchased most often was for domestic or international travel. Sample data obtained are shown in the following table.

	Type of Flight	
Type of Ticket	Domestic	International
First class	29	22
Business class	95	121
Economy class	518	135

a. Using a .05 level of significance, is the type of ticket purchased independent of the type of flight? What is your conclusion?
b. Discuss any dependence that exists between the type of ticket and type of flight.

12. A Deloitte employment survey asked a sample of human resource executives how their company planned to change its workforce over the next 12 months (*INC. Magazine,* February 2012). A categorical response variable showed three options: The company plans to hire and add to the number of employees, the company plans no change in the number of employees, or the company plans to lay off and reduce the number of employees. Another categorical variable indicated if the company was private or public. Sample data for 180 companies are summarized as follows.

	Company	
Employment Plan	**Private**	**Public**
Add Employees	37	32
No Change	19	34
Lay Off Employees	16	42

a. Conduct a test of independence to determine if the employment plan for the next 12 months is independent of the type of company. At a .05 level of significance, what is your conclusion?

b. Discuss any differences in the employment plans for private and public companies over the next 12 months.

14. A vehicle quality survey asked new owners a variety of questions about their recently purchased automobile (J.D. Power and Associates, March 2012). One question asked for the owner's rating of the vehicle using categorical responses of average, outstanding, and exceptional. Another question asked for the owner's education level with the categorical responses some high school, high school graduate, some college, and college graduate. Assume the sample data below are for 500 owners who had recently purchased an automobile.

	Education			
Quality Rating	**Some HS**	**HS Grad**	**Some College**	**College Grad**
Average	35	30	20	60
Outstanding	45	45	50	90
Exceptional	20	25	30	50

a. Use a .05 level of significance and a test of independence to determine if a new owner's vehicle quality rating is independent of the owner's education. What is the *p*-value and what is your conclusion?

b. Use the overall percentage of average, outstanding, and exceptional ratings to comment upon how new owners rate the quality of their recently purchased automobiles.

15. *The Wall Street Journal* Corporate Perceptions Study 2011 surveyed readers and asked how each rated the quality of management and the reputation of the company for over 250 world-wide corporations. Both the quality of management and the reputation of the company were rated on an excellent, good, and fair categorical scale. Assume the sample data for 200 respondents below applies to this study.

	Reputation of Company		
Quality of Management	**Excellent**	**Good**	**Fair**
Excellent	40	25	5
Good	35	35	10
Fair	25	10	15

a. Use a .05 level of significance and test for independence of the quality of management and the reputation of the company. What is the *p*-value and what is your conclusion?

b. If there is a dependence or association between the two ratings, discuss and use probabilities to justify your answer.

17. The National Sleep Foundation used a survey to determine whether hours of sleep per night are independent of age (*Newsweek*, January 19, 2004). A sample of individuals was asked to indicate the number of hours of sleep per night with categorical options: fewer than 6 hours, 6 to 6.9 hours, 7 to 7.9 hours, and 8 hours or more. Later in the survey, the individuals were asked to indicate their age with categorical options: age 39 or younger and age 40 or older. Sample data follow.

	Age Group	
Hours of Sleep	**39 or younger**	**40 or older**
Fewer than 6	38	36
6 to 6.9	60	57
7 to 7.9	77	75
8 or more	65	92

a. Conduct a test of independence to determine whether hours of sleep are independent of age. Using a .05 level of significance, what is the *p*-value and what is your conclusion?

b. What is your estimate of the percentages of individuals who sleep fewer than 6 hours, 6 to 6.9 hours, 7 to 7.9 hours, and 8 hours or more per night?